

FAIL

EN 1993-1-1 (UK NA) — unrestrained beam (LTB) member check 457 191 82 UB S275

Governing: **LTB+torsion (EN 1993-6 Annex A) = 2.006** (ULS: 1.35G + 1.5Q (Eq 6.10))

Shear 0.16 • Bending 0.65 • LTB 0.73 • Defl 0.99

Member Loading and Member Forces

457 191 82 UB S275 ($p_y = 275 \text{ N/mm}$) $L = 10 \text{ m}$

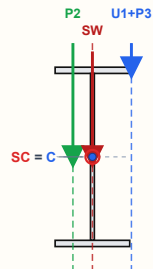
Self-weight 0.8044 kN/m ? $0 \text{ } 10 \text{ m}$ [Dead (G)] [automatic]

UDL 12 kN/m ? $0 \text{ } 10 \text{ m}$ [Imposed (Q)] [$e=100 \text{ mm}$, $z_g=228 \text{ mm}$]

Point 20 kN ? @ 4 m [Imposed (Q)] [$e=-50 \text{ mm}$, $z_g=0 \text{ mm}$]

Point 10 kN ? @ 6 m [Imposed (Q)] [$e=100 \text{ mm}$, $z_g=228 \text{ mm}$]

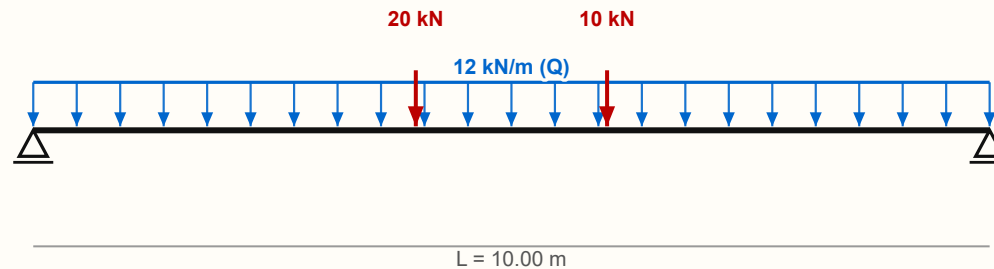
SECTION LOAD LINES



- SW: $e+0$, $z_g+0 \text{ mm}$
- U1+P3: $e+100$, $z_g+228 \text{ mm}$
- P2: $e-50$, $z_g+0 \text{ mm}$

3 load lines (same e/z_g grouped) • SC •
C • [click to enlarge](#)

LOADING



Reactions (governing-moment combo, ULS: 1.35G + 1.5Q (Eq 6.10)): $R@0\text{m} = 119.43 \text{ kN}$ $R@10\text{m} = 116.43 \text{ kN}$

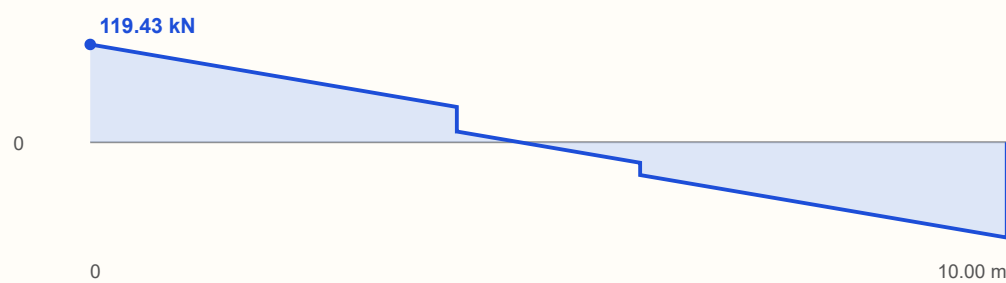
Load Combinations Considered

ULS combination	Max F_v (kN)	Max M_x (kN m @ m)	Governs
ULS: 1.35G + 1.5Q (Eq 6.10)	119.43	329.52 @ 4.69	governs shear, governs moment

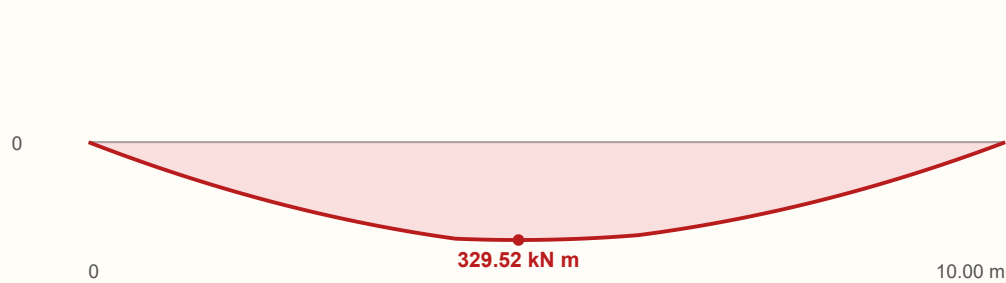
SLS combination	Deflection (mm)	Governs
SLS: Variable actions only (NA 2.23)	-27.63	governs deflection

Member Force Envelope ULS combinations Deflection SLS combinations			
Axial Force (kN)	Max Shear F _v (kN)	Max Moment M _x (kN m @ m)	Max Deflection (mm @ m)
200.000 C	119.43	329.52 @ 4.69	-27.63 @ 5

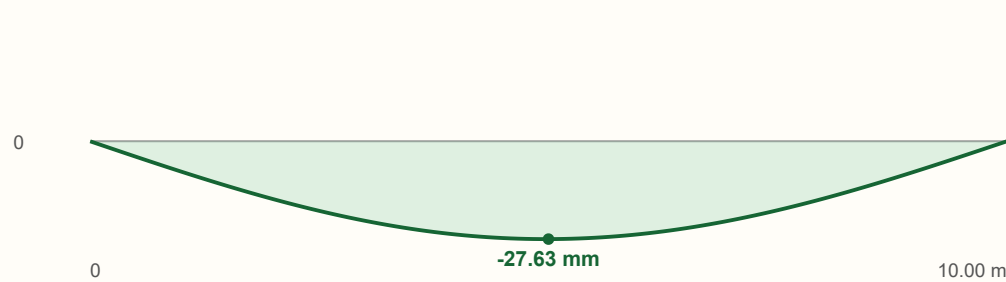
SHEAR FORCE (KN)



BENDING MOMENT (KN M)

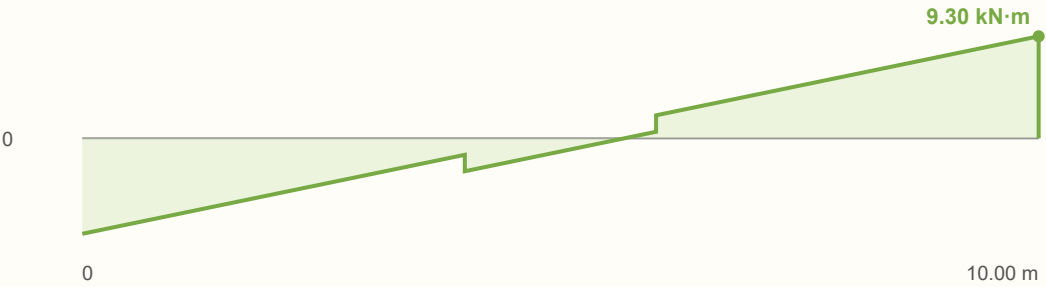


DEFLECTION (MM, SLS)



Bending-moment diagram plotted on the tension (sagging-down) side. Shear and deflection plotted to true sign (down = below the axis).

TORSIONAL MOMENT (KN·M) — PER-LOAD SHEAR-CENTRE ECCENTRICITIES (ULS: 1.35G + 1.5Q (EQ 6.10))



All supports are assumed to prevent twist (fork supports); the torque diagram is drawn for the combination governing torsion.

Classification of Cross-Section (EN 1993-1-1 Table 5.2)

Section (82 kg/m)	457 191 82 UB [S275] h=460, b=191.3, t _w =9.9, t _f =16, r=10.2 mm		
Nominal yield strength f _y (Table 5.1 / BS EN 10025-2)	t _f = 16 mm (t ≤ 16)	275 N/mm²	f _u = 410
ε = √(235/f _y)	√(235/275)	0.92	
Outstand flange: c = (b – t _w – 2r)/2	(191.3 – 9.9 – 2×10.2)/2	80.5 mm	
c/t _f = 5.03	Class 1 limit: 9ε = 9×0.92 = 8.32	5.03 < 8.32	Class 1
Web: c = d = 407.6 mm; c/t _w = 41.2	bending + compression (Table 5.2): α = 0.59, ψ = -0.86; limits 59.4ε / 68.4ε / 108.8ε	41.2 < 54.88	Class 1
Section class under pure bending	max(flange, web)		Class 1

Shear Resistance (Cl. 6.2.6)

A _v = A – 2bt _f + (t _w +2r)t _f	10400 – 2×191.3×16 + (9.9+2×10.2)×16	4763 mm²	
but not less than ηh _w t _w (η=1.0)	1.0×428×9.9 = 4237 mm²	A _v = 4763 mm²	4763 > 4237
V _{c,Rd} = V _{pl,Rd} = A _v (f _y /√3)/γ _{M0}	4763×(275/√3)/1.0 ×10 ⁻³	756 kN	
V _{Ed} / V _{c,Rd} ≤ 1.0	119.4 / 756	0.16	OK

Shear Buckling (Cl. 6.2.6(6))

h _w /t _w ≤ 72ε/η	428/9.9 = 43.2 vs 72×0.92/1.0 = 66.6	43.2 < 66.6	Not required
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Torsion Analysis — SCI P385 Method B (elastic; fork ends, warping free)

Torsional bending constant a = √(EI _w /GI _T)	I _T = 69.2 cm ⁴ ; I _w = 0.922 dm ⁶ (P385 App A)	a = 1.86 m; L/a = 5.38	
T _{Ed} (max. internal)	torque diagram by statics (fork ends)	9.30 kN·m	
Max rotation φ (ULS)	Cases 3/4/10 closed forms, superposed per combination	0.3061 rad = 17.54°	
Warping flange moment M _{w,Ed} = EI _w φ''/(h–t _f)	max over span	13.88 kN·m	
Rotation-induced M _{z,Ed} = φ·M _{y,Ed}	max coincident value (mandatory, P385)	100.37 kN·m	
End torque reactions T _t (for connection design)	x = 0 / x = L	5.50 / 5.87 kN·m	

Bending + Torsion Cross-Section (P385 §3.1.2 — plastic, Class 1/2)

Resistances	M _{pl,y,Rd} = 503; M _{pl,z,Rd} = 84; M _{pl,f,Rd} = W _{pl,z} /2·f _y = 41.8 kN·m		
(M _y /M _{pl,y})² + M _w /M _{pl,f} + M _z /M _{pl,z}	@ x = 5.15 m (ULS: 1.35G + 1.5Q (Eq 6.10)): M _y =327.5, M _w =12.39, M _z =100.23 kN·m	1.92	FAIL

Shear + Torsion (Cl. 6.2.7(9), Eq 6.26)

τ _{t,Ed} at governing V-T point	@ x = 0 m (ULS: 1.35G + 1.5Q (Eq 6.10)): V = 119.4 kN, T _t = 5.50 kN·m	78.7 N/mm²	
V _{pl,T,Rd} = √(1–τ _t /(1.25·f _y /√3))·V _{pl,Rd}	coincident sweep; V _{pl,Rd} = 756 kN	587 kN	
V _{Ed} / V _{pl,T,Rd}	119.4 / 587	0.2	OK

Moment Resistance (Cl. 6.2.5)

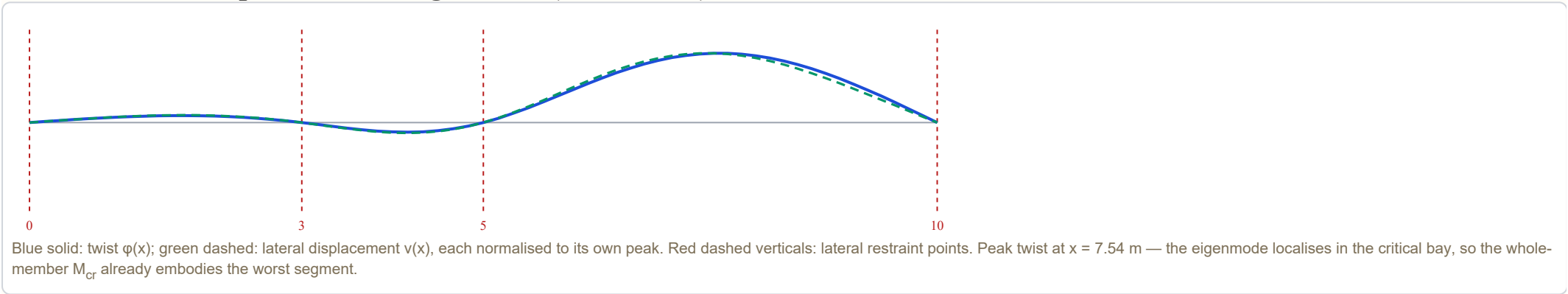
Shear at point of max. moment (Cl. 6.2.8)	$V_{Ed} @ x=4.69 \text{ m: } 0.1 \text{ kN vs } 0.5V_{pl,T,Rd} = 294 \text{ kN (cl 6.2.8(4))}$	no reduction Low shear
$M_{c,Rd} = M_{pl,Rd} = W_{pl,y} \cdot f_y / \gamma_{M0}$	$1830 \times 275 / 1.0 \times 10^{-3}$	503 kN·m
$M_{y,Ed} / M_{c,Rd} \leq 1.0$	330 / 503	0.65 OK

Lateral–Torsional Buckling — Elastic Critical Moment (FE eigenvalue solution)

Buckling model	FE eigenvalue solution of $(K_e + \lambda K_g)d = 0$ over the governing BMD; Hermite cubics, 4 DOF/node (v, v', ϕ, ϕ'); 63 elements, Richardson-extrapolated	mesh err < 0.001 %
Section properties	$I_z = 1870 \text{ cm}^4$; $I_T = 69.2 \text{ cm}^4$; $I_w = 0.922 \text{ dm}^6$; $G = 81000 \text{ N/mm}^2$	$z_j = 0 \text{ mm}$
Lateral restraints	2 support(s) taken as fork restraints ($v = \phi = 0$); 2 intermediate restraint(s)	—
Load height	per-load $z_g = +228, 0 \text{ mm}$ above the shear centre; refs: top +230 mm, bottom -230 mm (max value destabilising)	$M_{cr} \text{ (load reversed)} =$
M_{cr}	eigenvalue $\times \max M(x) $ — no C_1, C_2 or C_3 used	1145.2 kN·m
$\tilde{\lambda}_{LT} = \sqrt{(W_y f_y / M_{cr})}$	$\sqrt{(1830 \times 10^3 \times 275 / 960.6 \times 10^6)}$	960.6 kN·m
Buckling curve	NA Table 6.3, $h/b = 2.4$	0.724
Φ_{LT} ; χ_{LT}	$\tilde{\lambda}_{LT,0} = 0.4, \beta = 0.75 \text{ (NA 2.17)}; \Phi = 0.776$	curve c ($\alpha_{LT} = 0.49$)
C_1 (for k_c only)	back-calculated from the eigen solution: $M_{cr}(z_g=0, z_j=0) / M_{cr}(\text{uniform moment, same restraints})$	$\chi_{LT} = 0.811$
$k_c = 1/\sqrt{C_1}$; $f = 1 - 0.5(1 - k_c)[1 - 2(\tilde{\lambda}_{LT} - 0.8)^2] \leq 1$	$k_c = 0.803 \text{ (NA 2.18)}$	$C_1 = 1.552$
$\chi_{LT,mod} = \chi_{LT} / f \leq \min(1, 1/\tilde{\lambda}^2)$	0.811/0.903	$f = 0.903$
$M_{b,Rd} = \chi_{LT,mod} W_{pl,y} f_y / \gamma_{M1} \leq M_{c,Rd}$	$0.899 \times 1830 \times 275 / 1.0$	0.899
$M_{Ed} / M_{b,Rd}$	329.5 / 452.3	452.3 kN·m
		0.73 OK

• Minor-axis strut buckling length $L_{cr,z} = 5.00 \text{ m}$, taken as the largest spacing between adjacent lateral restraint points (SCI P360 6.2: secondary members act as bracing points; $k = 1.0$ between restraints). Ensure each restraint really is an effective bracing point - adequate stiffness, strength and anchorage. The major axis keeps $L_{cr,y} = L_E \times L = 10.00 \text{ m}$.

Buckled Mode Shape — critical eigenmode (normalised)



Segment-Method Comparison (informational — NOT the design basis)

Segment 0–3 m (isolated, fork ends)	$M_{\text{seg,Ed}} = 272.4$; $M_{\text{cr,seg}} = 1702$; $M_{\text{b,Rd}} = 462 \text{ kN}\cdot\text{m}$; $\bar{\lambda}_{\text{LT}} = 0.544$; $\chi_{\text{LT}} = 0.919$ (no f-factor)	util = 0.589
Segment 3–5 m (isolated, fork ends)	$M_{\text{seg,Ed}} = 329.5$; $M_{\text{cr,seg}} = 2319$; $M_{\text{b,Rd}} = 485 \text{ kN}\cdot\text{m}$; $\bar{\lambda}_{\text{LT}} = 0.466$; $\chi_{\text{LT}} = 0.963$ (no f-factor)	util = 0.68
Segment 5–10 m (isolated, fork ends)	$M_{\text{seg,Ed}} = 328.6$; $M_{\text{cr,seg}} = 583$; $M_{\text{b,Rd}} = 344 \text{ kN}\cdot\text{m}$; $\bar{\lambda}_{\text{LT}} = 0.929$; $\chi_{\text{LT}} = 0.683$ (no f-factor)	util = 0.956 worst segment

Conventional check for comparison: each bay between lateral restraints is isolated with fork ends and its own share of the moment diagram and loads. Isolation discards the lateral-bending (v') and warping (φ') continuity that the adjacent bays provide across the restraint points, so these values are conservative. The design basis is the whole-member eigen solution above: $M_{\text{cr}} = 960.6 \text{ kN}\cdot\text{m}$ vs the worst isolated segment $583 \text{ kN}\cdot\text{m}$.

LTB + Torsion Interaction (BS EN 1993-6 Annex A / P385 §6.2)

Basis	$M_{\text{b,Rd}} = \chi_{\text{LT}} W_y f_y$ (no f-factor) = $408 \text{ kN}\cdot\text{m}$; $M_{\text{cr}} = 961 \text{ kN}\cdot\text{m}$	$C_{\text{mz}} = 1$; $k_\alpha = 1.52$
$M_y/M_{\text{b,Rd}} + C_{\text{mz}} M_z/M_{\text{pl,z}} + k_w k_{\text{zw}} k_\alpha M_w/M_{\text{pl,f}}$	@ x = 4.94 m: $328.9/408 + 1 \times 100.33/83.6 + 0.64 \times 0 \times 1.52 \times 11.99/41.80$	2.01 FAIL

Cross-Section Axial + Bending (Cl. 6.2.9)

$N_{\text{pl,Rd}} = A f_y / \gamma_{\text{M0}}$	$104 \text{ cm}^2 \times 275$	2860.0 kN
$N_{\text{Ed}} / N_{\text{pl,Rd}}$	$200.0 / 2860.0$; n = 0.07	0.07 OK
$M_{\text{N,y,Rd}}$ (cl 6.2.9.1)	a = 0.411; $\alpha = 2$, $\beta = 1$; small axial (cl 6.2.9.1(4)): no major reduction	503.25 kN·m
$(M_{\text{y,Ed}}/M_{\text{N,y,Rd}})^\alpha + (M_{\text{z,Ed}}/M_{\text{N,z,Rd}})^\beta$	$(329.52/503.25)^2 + (10.00/83.60)^1$	0.548 OK

Member Buckling Resistance (Cl. 6.3.3, Annex B Method 2 — Table B.2)

$L_{\text{cr,y}} = 1 \cdot L$; $L_{\text{cr,z}}$ = restraint spacing (P360 6.2)	$\bar{\lambda}_y = 0.613$ (curve a); $\bar{\lambda}_z = 1.362$ (curve b)	10 / 5 m
$N_{\text{b,y,Rd}}$; $N_{\text{b,z,Rd}} = \chi A f_y / \gamma_{\text{M1}}$	$\chi_y = 0.885$; $\chi_z = 0.398$	2531.4 / 1139.3 kN
$C_{\text{my}} = C_{\text{mLT}}$ (Table B.3)	mixed loading (larger of uniform/concentrated), $\alpha_h = 0.000$, $M_h = 0.0$, $M_s = 328.6$, $\psi = 0.00$	0.95; $C_{\text{mz}} = 1$
k_{yy} ; k_{zz} ; k_{yz} ; k_{zy}	Table B.2 (susceptible): $1 - 0.1 \cdot \min(\bar{\lambda}_z, 1) \cdot n_z / (C_{\text{mLT}} - 0.25)$	0.981; 1.246; 0.747; 0.975
$N_{\text{Ed}}/N_{\text{b,y,Rd}} + k_{yy} M_{\text{y,Ed}}/M_{\text{b,Rd}} + k_{yz} M_{\text{z,Ed}}/M_{\text{c,z,Rd}}$ (Eq 6.61)	$0.079 + 0.981 \times 329.52/452.27 + 0.747 \times 10.00/83.60$	0.883 OK
$N_{\text{Ed}}/N_{\text{b,z,Rd}} + k_{zy} M_{\text{y,Ed}}/M_{\text{b,Rd}} + k_{zz} M_{\text{z,Ed}}/M_{\text{c,z,Rd}}$ (Eq 6.62)	$0.176 + 0.975 \times 329.52/452.27 + 1.246 \times 10.00/83.60$	1.035 FAIL

Vertical Deflection of Beam (BS EN 1993-1-1 NA 2.23 — SLS: Variable actions only (NA 2.23))

w_{max} (governing SLS combination)	@ x = 5 m	27.6 mm
Limit = span/360	$10000/360 = 27.8 \text{ mm}$	$27.6 < 27.8 \text{ mm}$ OK

Rotation at SLS (SLS: Variable actions only (NA 2.23))

Max rotation φ_{ser}	@ x = 5.23 m; P385 practical guide: $\varphi \leq 2^\circ$ (no codified limit — advisory)	$0.2041 \text{ rad} = 11.69^\circ > 2^\circ$ — review
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ADVISORY (does not block PASS): Web bearing and buckling of the unstiffened web under concentrated loads and at supports (EN 1993-1-5 clause 6) are outside this calculator's scope - verify separately wherever a point load or a reaction is applied to the web.

ADVISORY (does not block PASS): Serviceability twist $\theta_{\max} = 11.7^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

Open-section torsion per SCI P385: elastic Method B (fork ends both supports, warping free), Appendix C Cases 3/4/10 closed forms superposed per combination; effects evaluated coincidentally along the span. Cross-section verified by the P385 §3.1.2 interaction; shear by Eq 6.26; buckling by the BS EN 1993-6 Annex A bending–torsion interaction with k_α amplification. Plastic-design caution: φ used for $M_{z,\text{ind}}$ is the elastic value (P385 notes rotation may exceed it when plastic resistance is mobilised). Destabilising load-height growth of e (P385 §2.3.3) is not iterated — if loads sit above the shear centre, add an allowance to e or use the destabilising switch.

Torsion: each applied load acts at its own offset e from the shear centre; loads with $e = 0$ and applied moments generate no torque. Automatic self-weight acts through the centroid, so it has $e = 0$ for doubly symmetric sections but $e = e_{\text{sc}}$ for PFC channels. The verification conservatively assumes maximum shear, bending and torsion are coincident (SCI example note). All supports are assumed to prevent twist — fork supports; friction-grip connections or similar may be required. Torsion is evaluated on the EC3 code path only.

Unrestrained beam: design follows BS EN 1993-1-1 (UK NA) with M_{cr} solved directly by the finite-element eigenvalue method over the governing moment diagram. Load height z_g , mono-symmetry z_j , lateral restraints, and cantilever root warping are included in M_{cr} ; no SN003a/SN006a C-table or P362 simplified slenderness route is used.

Design basis for the LTB verdict: elastic critical moment from the finite-element eigenvalue solution (4 DOF/node: v, v', φ, φ' ; z_g and z_j included; 63 elements, Richardson-extrapolated).

Axial + bending per EN 1993-1-1: cross-section by cl 6.2.9 (plastic $M_{N,Rd}$, Class 1/2); member buckling by cl 6.3.3 with Annex B Method 2 interaction factors (Table B.2 — susceptible, C_m per Table B.3 from the governing moment diagram). Strut lengths: $L_{\text{cr},y} = L_{\text{E-factor}} \times L$; $L_{\text{cr},z}$ = largest lateral-restraint spacing (SCI P360 6.2). The destabilising $\times 1.2$ switch is an LTB concept and is NOT applied to strut buckling. N_{Ed} is the direct design value (not run through the combinations). Validated against an independent commercial-software SHS beam-column worked example (C_m 0.4, k_{zy} 0.24, Eq 6.61 0.184, Eq 6.62 0.110).

χ_{LT} uses $\lambda_{LT,0}=0.4$, $\beta=0.75$ and buckling curve per NA 2.17 (Table 6.3: $h/b \leq 2 \rightarrow b$; $2 < h/b \leq 3.1 \rightarrow c$; $h/b > 3.1 \rightarrow d$); $\chi_{LT,\text{mod}} = \chi_{LT}/f$ with $k_c = 1/\sqrt{C_1}$ (NA 2.18), where C_1 is back-calculated from the shape-only eigen result for k_c only. The design strength f_y from the flange thickness is used consistently in every expression, including Eq 6.56.

Analysis: 2-node Euler Bernoulli beam elements (direct stiffness); reactions exact, shear/moment by statics, deflection at nodes exact. Section data: SCI P363 Blue Book. This is a design aid results to be verified by a competent engineer.